QQ邮箱 mail.qq.com

MMM 2027 - Submission Result Notification

发件人 MMM 2027 Organizing Committee <mmm2027@conftool.pro>
收件人 3231865020 <3231865020@qq.com>
抄送 bin25981 <bin25981@gmail.com>, jqz39688661 <jqz39688661@2925.com>, chenb5 <chenb5@stu.ujn.edu.cn>, ise_suj <ise_suj@ujn.edu.cn>, ise_zhangjing <ise_zhangjing@ujn.edu.cn>
大小 43K
时间 2026年10月19日 22:50

Dear bin chen,

Thank you for your submission to MMM 2027. This year we received a record number of submissions and the selection process was very competitive. All papers were reviewed by 3 - 5 members of the international review committee. We are delighted to inform you that your submission 'RoGS-SLAM: Robust Gaussian Splatting SLAM in Dynamic Environments via Reliability-Guided Weighting' was accepted as a full paper to MMM 2027 in the 'Reconstruction and Generation of Humans, Objects, and Scenes for Spacial Intelligence' special session. Please read this email carefully.

We encourage you to consider the comments and feedback of the reviewers when preparing your final version of your paper, that will be due to be uploaded on conftool (PDF paper, zipped source files and copyright form signed) by 1 November 2026.

At the MMM 2027 conference, accepted papers will be presented in either on-site oral, on-site poster, or online short video on our online platform. We will decide on your presentation format once we know if you intend to be physically present at the conference in Siem Reap, or if you intend to participate online. We will capture your preference in the registration system. Please note that all papers (onsite presented or online-only) are considered to be full-papers in the proceedings and that every accepted paper must be covered by a full conference registration.

Within a few days, we will send you details of the camera-ready paper upload process. Please pay attention to this email and follow the instructions it contains.

Once again, congratulations on getting accepted for publication at MMM 2027. Your reviews and feedback are included below.

Contribution Details

ID: 826
Title: RoGS-SLAM: Robust Gaussian Splatting SLAM in Dynamic Environments via Reliability-Guided Weighting

Review Result of the Program Committee: This contribution has been accepted for presentation at MMM 2027.

Overview of Reviews

Review 1
========

Contribution of the Submission

The paper presents RoGS-SLAM, an extension of RGB-D MonoGS for dynamic environments. It combines temporally bounded keyframe insertion, robust residual reweighting, flow-and-depth-based reliability weighting, and a confirmation-based Gaussian lifecycle. An optional person mask defines a separate combined configuration. The main contribution is an empirical analysis of regime-dependent component effects, an equal-flow-input comparison against hard masking, and a pose-error study showing different responses of local and full-trajectory metrics. The paper explicitly reports failure cases and limits its accuracy and deployment claims.

Evaluation of the Contribution

Quality of Content (10%): 6
Significance (10%): 6
Originality (10%): 6
Thematic Relevance (10%): 10
Presentation (10%): 6
Overall Recommendation (50%): 7
Total points (out of 10) : 6.9

Comments for the Authors

This paper investigates robust Gaussian-splatting SLAM in dynamic RGB-D scenes. Its principal contribution is an empirical examination of when the proposed components help, rather than a claim of a fundamentally new estimation primitive or state-of-the-art accuracy.

Several aspects are valuable. The paper distinguishes the mask-free and combined configurations, reports negative results, and acknowledges the limitations of static-scene performance and majority-based reliability statistics. The three-seed experiments, factorial design, and equal-flow-input comparison are useful steps toward understanding the system. The distinction between local relative pose error and accumulated trajectory error is also worth studying.
Major concerns
1.The empirical contribution needs a more accessible and auditable result package.

Sections 4 and 5 repeatedly refer to pre-registered decisions, run directories, campaign-specific anchors, and noise floors. However, the submitted manuscript does not provide an accessible registration record, configuration archive, or complete per-sequence result table.

Please provide the exact sequence lists for the 18-sequence evaluation, pilot sets, factorial experiments, and held-out checks. Include all eight factorial cells, per-seed results, completion status, and a mapping between reported results and configurations. The provenance of the pre-registered rules and subsequent deviations should also be documented.

This is especially important because Section 5.3 reports substantial variation between repeated runs and campaigns. The manuscript appropriately acknowledges this variation, but readers need the underlying records to assess the stability of the conclusions.

2.Several central method definitions are incomplete.

Section 3.4 describes how flow disagreement is normalized into a bounded cue, but does not give the complete mapping. The depth-residual anomaly, opacity gating, and coefficient v are also insufficiently specified. Section 3.5 introduces Gaussian support and contradiction evidence without defining precisely how a view supplies either type of evidence.

Please provide the complete equations or pseudocode, all configuration constants, visibility and correspondence rules, and the treatment of invalid depth and occlusions. Specify whether the robust and reliability weights multiply the residual, squared residual, or loss term. This distinction changes the effective optimization objective.

The warm-up timing, pose used to compute rigid flow, and Gaussian lineage operations should be explicit enough to reproduce without access to an unspecified run configuration.

The flow comparison supports a bounded conclusion, but does not fully isolate the weighting mechanism.

3.Section 5.4 makes a useful comparison using identical frozen flow. However, the hard-mask counterpart also lacks the MAD normalization and geometric cue. The paper acknowledges this, which is appropriate. Nevertheless, several changes separate the two treatments.

Please document which keyframe, robust-loss, and Gaussian-lifecycle settings are identical across the three arms. A particularly informative control would retain the same normalized flow-and-depth score and vary only hard rejection versus soft weighting. Separately enabling reliability at tracking and lifecycle confirmation would help identify its two contributions.

The present experiment supports the advantage of the proposed treatment over the specified naive counterpart. Broader mechanism attribution should remain limited unless these additional differences are controlled.

-=-

Review 2
========

Contribution of the Submission

This paper presents RoGS-SLAM, a robust 3D Gaussian Splatting-based SLAM framework for dynamic environments. The proposed approach introduces a pixel-level reliability weighting mechanism based on optical flow consistency and geometric residuals to reduce the influence of dynamic objects during camera tracking. It also incorporates robust residual reweighting, keyframe management, and Gaussian lifecycle control. The authors conduct extensive experiments on dynamic RGB-D benchmarks, including comparisons with existing SLAM systems, component ablations, and controlled evaluations of the reliability weighting strategy. The main contribution lies in the integration and systematic evaluation of reliability-guided estimation techniques for improving the robustness of Gaussian Splatting SLAM in dynamic scenes.

Evaluation of the Contribution

Quality of Content (10%): 8
Significance (10%): 8
Originality (10%): 6
Thematic Relevance (10%): 8
Presentation (10%): 8
Overall Recommendation (50%): 7
Total points (out of 10) : 7.3

Comments for the Authors

This paper presents RoGS-SLAM, a robust Gaussian Splatting-based SLAM framework designed to improve camera tracking in dynamic environments. The authors introduce a reliability-guided weighting mechanism and integrate several techniques for robust estimation, keyframe management, and Gaussian map maintenance. The topic is relevant to dynamic SLAM and 3D scene reconstruction.

Strengths:
1. This paper addresses an important and practical problem in dynamic SLAM. The proposed reliability-guided weighting strategy is well motivated and provides a reasonable alternative to directly excluding potentially dynamic pixels.

2. The experimental evaluation is relatively comprehensive. The authors compare the proposed approach with several representative SLAM systems and conduct extensive ablation studies across different dynamic scenarios. The use of multiple random seeds and controlled comparisons strengthens the credibility of the experimental results.

3. The reported improvements in challenging dynamic scenes are encouraging. In particular, the proposed system substantially reduces trajectory errors compared with the original backbone in crowded environments.

4. The paper provides a relatively transparent discussion of the limitations and failure cases, which helps clarify the applicability of the proposed approach.

Weaknesses:

1. Several components of the proposed framework, including robust residual weighting and keyframe management, are based on established techniques. The authors could further clarify the specific technical novelty of the reliability-guided weighting mechanism and its differences from existing dynamic SLAM approaches.

2. Although the mask-free configuration is an important aspect of the framework, its performance is considerably weaker than the combined configuration on crowded sequences. A more detailed discussion of when semantic information is necessary would help clarify the practical advantages and limitations of the proposed method.

3. The ablation experiments are extensive, but the independent contribution of the Gaussian lifecycle mechanism could be investigated further. This would provide a more complete understanding of the contribution of each component.

4. The reported processing speed is relatively low, and optical flow estimation is excluded from the online runtime measurement. A more complete analysis of the end-to-end computational cost would be useful. In addition, further investigation of the sequences where the proposed approach underperforms the baseline could help improve the robustness of the system.

-=-

Review 3
========

Contribution of the Submission

The main contribution of this submission is the proposal of ROGS-SLAM, a mask-free core mechanism for dynamic 3DGS-SLAM that handles dynamic objects by measuring the disagreement between observed optical flow and rigid scene reprojection flow to down-weight unreliable pixels, rather than completely discarding them via semantic segmentation. Furthermore, the authors demonstrate through rigorous pre-registered factorial experiments that the necessity of different system components is highly regime-dependent, meaning no single bundle is universally necessary across all sequences. They also prove via strictly controlled experiments that the tracking performance gain is driven by their specific reliability-guided weighting mechanism rather than merely the introduction of external frozen optical flow information. Finally, the paper exposes a variance-bias trade-off in the pose estimation phase, highlighting that pose-side claims must explicitly specify their evaluation endpoints.

Evaluation of the Contribution

Quality of Content (10%): 6
Significance (10%): 6
Originality (10%): 6
Thematic Relevance (10%): 8
Presentation (10%): 6
Overall Recommendation (50%): 7
Total points (out of 10) : 6.7

Comments for the Authors

Strengths:
1. Exceptional Methodological Rigor: The experimental design is a benchmark for SLAM system papers. The use of a pre-registered 120-run factorial ablation to demonstrate that component necessity is "regime-dependent" is highly convincing. Furthermore, controlling for the exact same byte-identical frozen flow to prove the superiority of the weighting mechanism over a naive hard threshold is a brilliant and scientifically sound approach.
2. Scientific Transparency: The authors demonstrate high academic integrity by explicitly discussing the variance-bias trade-off in the pose-side mechanism. They also honestly report failure cases, including the system's degradation on purely static scenes and sequences where it performs worse than the unmodified backbone.
3. Clarity of Presentation: The writing is concise, direct, and free of unsubstantiated claims. The figures clearly illustrate the differences between the backbone and the proposed configurations.
Weaknesses & Suggestions for Improvement:
1. The "Mask-free" Claim and Computational Cost: While the system does not use semantic class priors, it heavily relies on a frozen RAFT network for dense optical flow. The authors state that the 54 ms/frame RAFT inference time is excluded from the online rate calculation. Since dense

optical flow is a significant computational bottleneck, excluding it makes the reported 0.75 FPS seem slightly disconnected from a true end-to-end deployment scenario.
  Suggestion: Please provide a brief discussion on the total end-to-end latency if the flow network were to run online, or clarify the feasibility of replacing RAFT with a more lightweight optical flow estimator.
2. Degradation in Static Environments: The paper notes in the Limitations section that the combined configuration performs worse and is less stable than the MonoGS backbone on static datasets like Replica.
  Suggestion: The authors should provide a deeper analysis of this degradation. Is it primarily caused by inherent noise in the RAFT flow predictions that incorrectly down-weights valid static geometry? Discussing the root cause would strengthen the paper.
3. Catastrophic Failure Cases: The paper mentions that on 4 out of 18 sequences, both proposed arms perform worse than the backbone, sometimes by an order of magnitude.
  Suggestion: Please elaborate on the characteristics of these specific sequences. What specific types of motion, occlusion, or texture patterns cause the reliability signal to fail so severely?
4. Generalization Across Backbones: The empirical evidence is currently tied strictly to the MonoGS RGB-D backbone.
  Suggestion: A brief discussion on whether this reliability-guided weighting mechanism could be easily integrated into other recent 3DGS SLAM architectures would enhance the perceived impact of the contribution.

-=-

Review 4
========

Contribution of the Submission

The paper studies robust RGB-D Gaussian-splatting SLAM in dynamic environments using a MonoGS backbone. It combines temporal keyframe coverage, Huber residual reweighting, a reliability signal derived from optical-flow and depth consistency, and a Gaussian confirmation/deletion lifecycle. It evaluates both a mask-free configuration and a combined configuration with semantic person masks. Its main contribution is an empirical analysis of component effects, a control using the same frozen optical flow, and the relationship between local pose errors and trajectory accuracy. The combined configuration reports 2.2-2.3 cm ATE on two BONN crowd sequences, alongside substantial limitations in other regimes.

Evaluation of the Contribution

Quality of Content   (10%): 6
Significance   (10%): 6
Originality   (10%): 10
Thematic Relevance   (10%): 4
Presentation   (10%): 4
Overall Recommendation (50%): 5
Total points (out of 10)   : 5.5

Comments for the Authors

The paper addresses a relevant problem in dynamic RGB-D SLAM. Its emphasis on understanding when components help, rather than claiming universal accuracy improvements, is valuable. However, several method definitions and experimental inferences need clarification.

Strengths

The combined configuration reports 2.3/2.2 cm ATE on crowd/crowd2, compared with 86.5/147.5 cm for the backbone. The three-seed experiments, component-removal analysis, and control using identical frozen flow are useful. The paper acknowledges regressions, the dependence on semantic masks in difficult scenes, and the exclusion of RAFT preprocessing from runtime. These disclosures help delimit the contribution.

Main concerns

1. Tracking objective and method specification.
Section 3.3 describes replacing an L2 tracking residual, but MonoGS [13] uses L1 photometric and geometric losses. Please specify the actual baseline and proposed objectives, including residual normalization and how the weights are applied. The definitions of e_flow, the geometric cue g, and

the lifecycle's support/contradiction tests also need to be made explicit.

2. Attribution in the controlled comparisons.
The same-flow control also changes the geometric cue and MAD normalization, so it does not isolate the benefit of soft weighting. A comparison using identical cues, with other components and optimization budgets held fixed, would strengthen attribution. Section 3.6 also appears to omit the added warm-up tracking mask from the simpler-to-combined change list.

3. Completeness of the empirical evidence.
Table 1 reports four BONN sequences, while the main conclusions draw on 18 sequences and additional held-out evaluations. Please provide the complete sequence lists, per-seed results and completion status, and full factorial and kernel/cue comparisons. The stated preregistration should be supported by an anonymized frozen protocol. Backend iteration and synchronization budgets are also important, given the substantial improvement on f3_st_hf when the asynchronous budget is increased.

The empirical focus is valuable, but clearer method definitions and more complete evidence are needed to assess the central conclusions reliably.

Best Regards
Your organizers of 33rd International Conference on MultiMedia Modeling.

--
MMM 2027 - 33rd International Conference on MultiMedia Modeling
January 05–08, 2027 | Siem Reap, Cambodia

https://www.conftool.pro/mmm2027/
https://mmm2027.net/